

# **(TR-104) PYTHON WITH AI/ML**

## **Training Day 32 Report**

**Date: 24 February 2026**

### **Topic Covered: Tokenization Internals, Attention Mechanics & Manual Transformer Inference**

The thirty-second day of training focused on understanding the internal working of transformer models including subword tokenization, attention mechanisms, hidden states, contextual embeddings, and manual transformer inference using Hugging Face models.

- Studied the limitations of traditional word-level tokenization and understood the need for subword tokenization in transformers.
- Explored different subword tokenization techniques including WordPiece, Byte Pair Encoding (BPE), and SentencePiece.
- Understood how transformers handle unknown and rare words using subword tokens.
- Loaded and analyzed pretrained tokenizers using the Hugging Face Transformers library.
- Inspected tokenizer outputs including `input_ids`, `token_type_ids`, and `attention_mask`.
- Converted input IDs back into readable tokens for better understanding of tokenization internals.
- Identified and understood the role of special tokens such as [CLS], [SEP], and padding tokens.
- Learned how padding and truncation work in transformer models for fixed-length input processing.
- Studied the importance of `attention_mask` and how it prevents the model from attending to padding tokens.
- Created padded inputs manually and analyzed masked token positions.
- Performed manual forward passes using transformer models without pipeline abstraction.
- Loaded transformer models with hidden states and attention outputs enabled.

- Inspected hidden state tensors generated at different transformer layers.
- Understood the shape and meaning of hidden state tensors in transformer architectures.
- Explored attention tensors and analyzed multi-head attention outputs.
- Performed attention weight analysis to observe token-to-token attention relationships.
- Interpreted attention behavior in contextual sentences for understanding semantic relationships.
- Studied layer-wise contextual representation evolution across transformer layers.
- Compared embeddings from embedding layers, middle layers, and final transformer layers.
- Conducted contextual embedding experiments using the word “bank” in different sentence contexts.
- Computed cosine similarity between contextual embeddings to observe semantic differences.
- Understood how transformers generate context-aware embeddings unlike static word embeddings such as Word2Vec.
- **GitHub Link:**  
<https://github.com/JaspinderKaurWalia26/Transformers>
- **Blog Link:**  
<https://medium.com/@jaspinderkaurwalia855/understanding-tokenization-attention-and-context-in-transformers-4196de2da1a2>

Overall, the training helped in strengthening understanding of transformer internals, subword tokenization, attention mechanisms, contextual embeddings, hidden states, and manual inference workflows using Hugging Face transformer models.